

Does a Mediterranean Diet Reduce First Cardiovascular Events in High-Risk Adults? A Single Retracted-and-Republished Trial Carries Most of the Randomised Evidence

Prepared with Claude (Anthropic)*

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Abstract

A Mediterranean dietary pattern is widely recommended for cardiovascular prevention, and that recommendation is usually traced to randomised evidence. We asked how strong the randomised evidence actually is for one specific question: whether a Mediterranean diet reduces *first* major cardiovascular events in adults at high cardiovascular risk who are free of cardiovascular disease at baseline. We searched OpenAlex, Crossref, Europe PMC and Semantic Scholar, verified every candidate against Crossref, and screened each for retraction and expressions of concern; 16 sources were retained. The primary-prevention question rests almost entirely on one trial, PREDIMED, whose 2013 report was retracted in 2018 after 1588 of 7447 participants were found not to have been individually randomised. In the republished reanalysis the effect persisted, with hazard ratios of 0.69 (95% CI 0.53–0.91) for a Mediterranean diet with extra-virgin olive oil and 0.72 (0.54–0.95) with nuts, and 0.69 (0.53–0.92) for the combined arms when the 1588 participants were excluded. Certainty is not uniform across outcomes: a Cochrane review rated the stroke reduction (HR 0.60, 0.45–0.80) moderate quality but found only low-quality evidence, compatible with no effect, for cardiovascular mortality (HR 0.81, 0.50–1.32) and all-cause mortality (HR 1.0, 0.81–1.24). We judge the evidence sufficient to support a recommendation but not to support confident causal or mortality claims, principally because it is one unreplicated trial in one country testing diet plus free food provision rather than dietary advice alone. The principal limitation of this review is that no source was obtained in full text; all findings rest on verified metadata and abstracts.

Keywords: Mediterranean diet; primary prevention; cardiovascular disease; randomised controlled trials; research integrity; evidence certainty

1 Introduction

Cardiovascular disease prevention guidance routinely endorses a Mediterranean dietary pattern, and the endorsement is typically presented as resting on randomised rather than merely observational evidence. That distinction matters, because dietary patterns are strongly confounded by the rest of a person's life: people who eat this way also tend to smoke less, move more, and be wealthier.

The randomised literature on Mediterranean diets is large, but most of it addresses surrogate outcomes such as lipids and blood pressure, or addresses patients who have already had a cardiovascular event. The narrower question of whether the diet prevents a *first* event in people at high risk is supported by a much smaller body of work, and that body of work has an unusual feature: its central trial was retracted and republished.

This review asks a single question. In adults at elevated cardiovascular risk but free of cardiovascular disease at baseline, does a Mediterranean dietary pattern reduce the incidence of first major cardiovascular

*This review was researched and drafted by an AI system under human direction. See Section 7.

events, and how strongly does the randomised evidence support whatever answer we give? Trials enrolling participants with established disease are treated throughout as indirect evidence and labelled as such. Our contribution is not new data but calibration: an explicit account of how much weight the randomised evidence can bear, including the findings that cut against a favourable conclusion.

Before searching, we specified four results that would refute a favourable answer: (F1) the pivotal trial’s effect collapsing once improperly randomised participants are removed; (F2) benefit appearing only on surrogate markers; (F3) systematic reviews grading certainty as low; and (F4) the evidence base amounting to a single trial in a single setting. As reported below, F1 did not hold, F2 and F3 partly held, and F4 held.

2 Search protocol and source verification

Searches were run on 2 September 2026 across four independent indexes: OpenAlex, Crossref, Europe PMC (which carries MEDLINE curation) and Semantic Scholar. Retrieval proceeded in rounds: keyword discovery; citation expansion forward and backward from the pivotal trial; a deliberately adversarial round searching for criticism, retraction and failure to replicate; a recency round; and a final round targeting work we expected to exist but had not found, which is how the successor trial and the data-integrity literature were located.

Every candidate was verified against Crossref for authoritative bibliographic metadata and screened for retraction using Crossref `update-to` records, OpenAlex `is_retracted`, and Europe PMC/MEDLINE status. Sixteen sources were retained; all sixteen resolved. Three carried a non-clean publication status, and those three turned out to be central to the answer rather than incidental to it.

Screening across three signals rather than one was not redundant. The Indo-Mediterranean Diet Heart Study is recorded by Crossref and Europe PMC as subject to an expression of concern, while OpenAlex reported `is_retracted = false`; conversely, OpenAlex flagged the PREDIMED retraction notice as itself retracted, an artefact of the update relationship between a notice and the article it retracts. Taken together these two cases suggest that single-index screening of this literature can fail in both directions, missing a real concern and manufacturing a false one, though we observed only these two instances and make no general claim about the indexes.

2.1 What this review did not obtain

No source was retrieved in full text. The pivotal trial and the Cochrane review are not open access, the publisher site returned an automated-traffic block, and the MEDLINE interface returned a CAPTCHA, which we did not attempt to circumvent. Every finding below therefore rests on verified bibliographic metadata, structured abstracts, and in one case a table fragment recovered through search indexing. Where a figure reaches us through another paper’s reading of a source rather than the source itself, we say so.

3 The randomised evidence base

3.1 One trial carries the question

The Cochrane review of Mediterranean-style diets included 30 randomised trials with 12,461 participants [13]. It updates an earlier 2013 review of primary prevention [12] and so, unlike that version, was written with the retraction already in view; its assessment of PREDIMED is based on the reanalysed data. Of the nine primary-prevention trials comparing a Mediterranean diet against no or minimal intervention, none reported clinical endpoints at all: no cardiovascular mortality, no total mortality, and no non-fatal myocardial infarction or stroke. Those trials reported blood pressure and lipids. The clinical-endpoint figures for primary prevention in that review come from a single trial, PREDIMED.

Combining this with our own searches, which found no other primary-prevention trial reporting hard endpoints, we infer that the question is effectively carried by one trial. This is an inference from an absence in our search rather than a proof that no such trial exists.

PREDIMED randomised 7447 adults aged 55–80 in Spain, at high cardiovascular risk but with no cardiovascular disease at enrolment, to a Mediterranean diet with free extra-virgin olive oil, a Mediterranean

diet with free mixed nuts, or advice to reduce dietary fat [6]. Median follow-up was 4.8 years. It is worth being precise about what was tested: not dietary advice alone, but a dietary pattern combined with free provision of specific foods. Any inference to a diet-advice-only intervention is an extrapolation beyond what the trial examined.

3.2 The retraction, and what the reanalysis changed

In 2017 a statistical screen of baseline-variable distributions across 5087 randomised trials flagged irregularities inconsistent with randomisation [1]; the trialists cite this analysis in their own republication. The subsequent investigation established that 1588 of the 7447 participants had not been individually randomised as the protocol specified. The 2013 report was retracted in 2018 [5, 7], and the problem propagated beyond the main report: a prespecified secondary-outcomes analysis in a different journal was separately retracted and republished [16].

The reanalysis is the pivot of this review, because it is where our pre-specified falsifier F1 was tested. In the republication, 288 primary endpoint events occurred: 96 in the olive oil arm (3.8%), 83 in the nut arm (3.4%), and 109 in the control arm (4.4%). Adjusting for baseline characteristics and propensity scores, the hazard ratios were 0.69 (95% CI 0.53–0.91) for olive oil and 0.72 (0.54–0.95) for nuts. Critically, when the 1588 participants without documented individual randomisation were excluded, leaving 5859, the estimates were 0.71 (0.52–0.97), 0.68 (0.49–0.95) and 0.69 (0.53–0.92) for the combined Mediterranean arms [6].

The effect did not collapse. F1 does not hold, and this is the strongest single argument available for taking the trial seriously despite its history. It should be weighed knowing that we read these figures from an indexed abstract and a recovered table fragment rather than the full report.

One feature of the reanalysis deserves comment. The headline estimates are described as an intention-to-treat analysis adjusted for baseline characteristics *and propensity scores* [6]. In a trial where randomisation had worked as intended, such adjustment would be unnecessary, because randomisation balances baseline characteristics in expectation. Its use here is a reasonable response to a known allocation problem, but we would note that it moves the analysis some distance toward the logic of an observational study, and that the resulting confidence intervals reflect sampling variability rather than uncertainty about the allocation mechanism itself. This is our reading of what the adjustment implies, not a criticism made by any source we retrieved.

3.3 Certainty varies sharply by outcome

Treating PREDIMED as a single undifferentiated result obscures the most important feature of the evidence, which is that certainty is not uniform across outcomes. Table 1 summarises the position.

Table 1: Certainty of the primary-prevention randomised evidence by outcome. Effect estimates and GRADE-style quality ratings are as reported by the Cochrane review [13]; the composite endpoint estimate is from the republished trial [6]. CI denotes confidence interval.

Outcome	Effect estimate (95% CI)	Certainty	Compatible with no effect?
Stroke	HR 0.60 (0.45–0.80)	Moderate	No
Composite major event	HR 0.69 (0.53–0.91) ^a	Not graded	No
Cardiovascular death	HR 0.81 (0.50–1.32)	Low	Yes
All-cause mortality	HR 1.0 (0.81–1.24)	Low	Yes
Non-fatal endpoints vs. minimal intervention	Not reported by any trial in this comparison		

^aMediterranean diet with extra-virgin olive oil versus control; the nut arm gave 0.72 (0.54–0.95).

Stroke is the outcome on which the evidence is strongest: a reduction from roughly 24 to 14 events per 1000, rated moderate quality. Mortality is where the evidence is weakest. For cardiovascular death and for all-cause mortality the confidence intervals include no effect, and Cochrane rated both low quality

[13]. A pooled analysis across four trials likewise found no significant effect on all-cause mortality, at OR 0.77 (0.51–1.15) [14].

This is a meaningful distinction that popular summaries tend to flatten. The claim “a Mediterranean diet reduces cardiovascular events” is better supported than the claim “a Mediterranean diet helps you live longer,” and the trial evidence does not currently establish the latter.

4 Findings that complicate the conclusion

4.1 The largest pooled estimate is not a primary-prevention result

A 2024 systematic review and meta-analysis of randomised trials reports a markedly stronger effect than PREDIMED alone: OR 0.52 (0.32–0.84) for major adverse cardiovascular events [14]. Taken at face value this looks like independent corroboration. It is not, for three reasons that compound.

First, the analysis pools four trials — Lyon, Indo-Mediterranean, PREDIMED and CORDIOPREV — of which only PREDIMED is primary prevention. Second, the heterogeneity for that headline estimate is $I^2 = 87\%$, which indicates the trials are not estimating a common effect. Third, one of the four pooled trials, the Indo-Mediterranean Diet Heart Study, has been subject to a formal expression of concern about the reliability of its data since 2005 [8, 15]. That concern has never been resolved. The Cochrane review, by contrast, excluded two secondary-prevention trials from its main analyses precisely because of published concerns about data reliability [13].

We therefore judge that this pooled estimate should not be read as an answer to the primary-prevention question, and that its apparent strengthening of the effect relative to PREDIMED is an artefact of what was pooled. This is our inference from the composition of the analysis; the review’s own authors present the estimate as supporting both primary and secondary prevention.

4.2 Indirect evidence from secondary prevention

The secondary-prevention trials are consistent in direction. CORDIOPREV randomised 1002 patients with established coronary heart disease in a single Spanish centre and found 87 primary events in the Mediterranean arm against 111 on a low-fat diet, with multivariable-adjusted hazard ratios ranging across models from 0.719 (0.541–0.957) to 0.753 (0.568–0.998) [4]. The Lyon Diet Heart Study, as reported by Cochrane, gave adjusted hazard ratios of 0.35 (0.15–0.82) for cardiovascular mortality and 0.44 (0.21–0.92) for total mortality in 605 post-infarction patients, rated low quality [3, 13].

Two cautions apply. CORDIOPREV’s composite is broader than PREDIMED’s, adding revascularisation and peripheral artery disease, so the effect sizes are not directly comparable. And these are patients with established disease; consistency of direction across prevention settings is reassuring but is not evidence about first events.

4.3 The evidence base is narrow, and a sex difference is unresolved

Both major trials were conducted in Spain. CORDIOPREV was single-centre and 82.5% male, and reported a benefit in men (HR 0.669, 0.489–0.915) with no difference detected among its 175 women [4]. Whether that reflects a real sex difference or an underpowered subgroup cannot be settled from the published analysis.

The point is sharpened by a systematic review that set out specifically to assess primary prevention in women and admitted both randomised trials and prospective cohorts to its eligibility criteria. It ultimately included sixteen studies, all of them prospective cohorts, covering 722,495 women [11]. No randomised trial met inclusion. A reviewer looking for randomised evidence on this question in women, in other words, found none.

That cohort evidence is far more precise than any trial: cardiovascular incidence HR 0.76 (0.72–0.81) and total mortality HR 0.77 (0.74–0.80). Precision is not the same as validity, and adherence-based exposure cannot separate the diet from the life around it. It is also worth noting that the cohorts did not find a significant stroke reduction in women (HR 0.87, 0.76–1.01), whereas stroke is the outcome on which the trial evidence is strongest. We report that discrepancy rather than resolving it; the designs and populations differ.

5 Discussion

5.1 How strong is the evidence, stated plainly

For the composite of myocardial infarction, stroke and cardiovascular death in high-risk adults without established disease, one good-sized randomised trial reports roughly a 30% relative risk reduction, and that estimate survived the removal of every participant whose randomisation could not be documented. Against an absolute event rate of 4.4% over 4.8 years in the control arm, this is a reduction of about one percentage point in absolute terms.

We judge this sufficient to support recommending the dietary pattern, which in any case carries little downside, and insufficient to support confident causal language or any mortality claim. The specific reasons are that the finding is unreplicated, geographically narrow, and attached to an intervention that included free food provision rather than advice alone.

The most common error in summarising this literature is to treat the strength of the composite result as though it transferred to mortality. It does not. The second most common is to treat pooled estimates as corroboration when the pooling mixes prevention settings and includes a trial under an unresolved expression of concern.

5.2 The retraction should change confidence, but less than it appears

It would be easy to draw either of two wrong conclusions from the retraction. The first is that PREDIMED is discredited; the reanalysis excluding the improperly randomised participants argues directly against that, and the trial remains the best evidence available. The second is that the retraction was a technicality now safely behind us; that understates it, because 21% of participants were affected, the same underlying problem forced a separate retraction in another journal [16], and the republication attracted formal published correspondence [2], whose content we were unable to obtain and therefore do not characterise.

The defensible position is intermediate. The trial's central estimate is probably close to right, and our confidence that it is exactly right should be lower than the confidence interval alone implies, because the interval does not price in the design irregularity. Reviews written by the trial's own investigators [10] are not independent sources on this point and we have not used them as such.

5.3 What would change our answer

The replication is registered and running. PREDIMED-Plus, testing an energy-reduced Mediterranean diet with physical activity, has published its design [9], and our searches on 2 September 2026 retrieved only secondary analyses — body composition, bone health, gut microbiota — and no primary cardiovascular outcome report. We state this as a search result rather than a fact about the world.

A null primary-outcome result from PREDIMED-Plus, or from any adequately powered trial outside Spain, would substantially weaken the conclusion drawn here, since the current answer rests on a single unreplicated trial. A positive result in a different population would move the composite finding from moderate to strong. Neither mortality claim will be settled by trials of this size; establishing a mortality effect would require either far larger trials or individual participant data pooling that excludes trials with unresolved integrity concerns.

6 Limitations of this review

The limitation that most constrains what can be concluded is access. No source was read in full text. The two most load-bearing documents, the republished trial and the Cochrane review, were available only as abstracts, and the sensitivity analysis on which our assessment of falsifier F1 turns was recovered as an indexed table fragment. Abstracts systematically omit the conditions that qualify a finding, so the risk that a stated caveat was missed is real and asymmetric: it runs toward overstating certainty.

Second, the Lyon effect estimates reach us through the Cochrane review's reading of that trial rather than the trial itself, and are attributed accordingly.

Third, this is a rapid review, not a systematic one. There was no protocol registration, no dual independent screening, no formal risk-of-bias instrument applied by us, and no PRISMA flow diagram.

Certainty ratings reported here are those assigned by the Cochrane authors, not ratings we generated. Searches were in English-language indexes.

Fourth, no meta-analysis was computed for this review. Every pooled figure quoted is attributed to the published analysis that produced it.

Fifth, the finding that no PREDIMED-Plus outcome paper exists is an absence in four indexes on one date, which is weaker than a positive finding and could be overturned by a publication we did not retrieve.

7 Conclusion

A Mediterranean dietary pattern, provided together with extra-virgin olive oil or nuts, probably reduces first major cardiovascular events in high-risk adults. The estimate is a hazard ratio near 0.7, it survived exclusion of every improperly randomised participant, and it is directionally consistent with secondary-prevention trials and with large cohort studies.

The evidence supporting that sentence is thinner than the diet's reputation suggests. It is one trial, in one country, retracted and republished, testing food provision rather than advice, with certainty that is moderate for stroke, ungraded for the composite, and low — and compatible with no effect — for both cardiovascular and all-cause mortality. The most-cited pooled estimate is stronger than PREDIMED but mixes prevention settings and includes a trial under an unresolved expression of concern. A systematic review seeking randomised evidence in women found none.

The honest summary is that the direction of effect is reasonably secure, the magnitude is uncertain, the mortality benefit is unestablished, and the finding is unreplicated. Recommending the diet is well justified; describing the question as settled is not.

Disclosure of AI involvement

This review was researched, analysed and drafted by Claude, an AI system made by Anthropic, under human direction. Literature discovery and citation verification used federated queries across OpenAlex, Crossref, Europe PMC and Semantic Scholar, with every reference checked against Crossref and screened for retraction before inclusion. No citation was written from model memory. Claims were recorded in a structured evidence ledger with provenance tags before any prose was written, and statements inferred rather than stated by a source are marked as inferences in the text. Access limitations are reported in Section 6. The work has not been peer reviewed.

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